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Good Science

Producing Reliable Insights that Matter in Practice



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<https://julianfrattini.github.io/>



JulianFrattini/**bda4sci**

Public

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Julian Frattini // about me

2013 - 2019

Bachelor & Master of Science

 Technical University of Munich,
Munich, Germany

2020 – 2025

Ph.D. Student

 Blekinge Institute of Technology,
Karlskrona, Sweden

2025 - 2029

Postdoctoral Researcher

 Chalmers University of Technology,
Gothenburg, Sweden

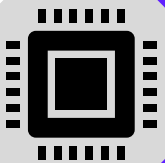


Julian Frattini // about my research



Good-enough Requirements Engineering

Quantifying the effect of quality factors of requirements artifacts on downstream software development activities



Future of Software Engineering

Paving the way for a future of software engineering driven by language-based AI agents.



Software Engineering Research Methodology

Improving the methods used in research to produce more rigorous and valid conclusions



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Disclaimer

Setting Your Expectations Right

But I am here to learn about

Software Engineering

and not

Software Engineering Research

Software Engineering

will become

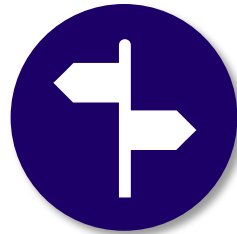
Software Engineering Research

Software Engineering will become Software Engineering Research

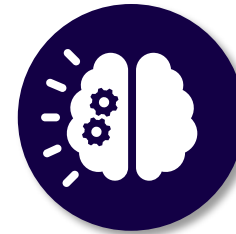
With the support of generative AI, traditional skills of software engineering (writing source and test code, designing an architecture, etc.) need to be complemented with



Communicating
with humans and LLMs
about expectations and
requirements



Making Decisions
about options that (much
faster) LLMs present to
you



Creating Knowledge
about the world and how
our software interacts
with it

You are exactly at the right place to learn about

Software Engineering

in the form of

Software Engineering Research



Motivation

Why Good Research is Worth the Effort

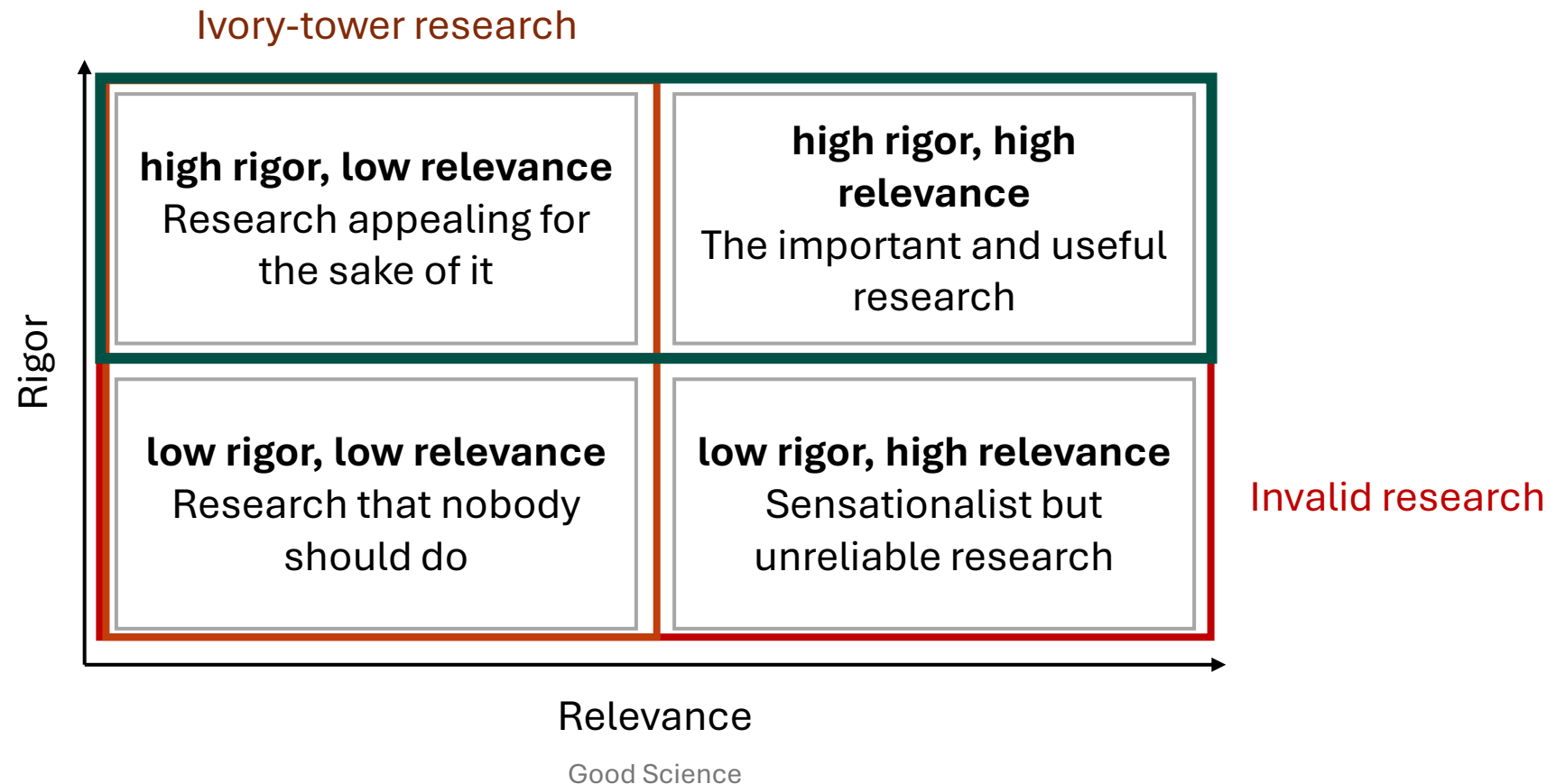
The Purpose of Science



” Good science informs decision-making policies.

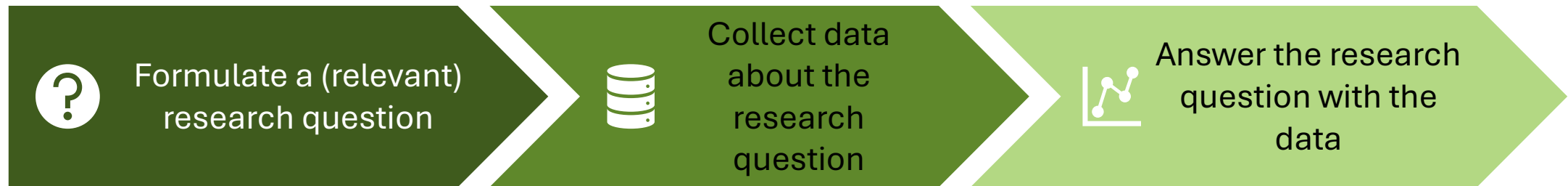
Properties of Science

What exactly constitutes “good science”?



The Scientific Process

One common type of the scientific process follows these steps:



For example: Does the use of artificial intelligence increase the productivity of software engineers?

The Scientific Process

When collecting data from a real scenario, we call this an **observational study** since we “observe” the reality. This type of study is very powerful, but underlies one critical challenge:



Is the observed effect *caused* by the variables of interest, or does it just *coincide* with other variables?

In our example, we might see that software engineers using AI are more productive. But is it *because* they use AI? Or is it because *motivated* engineers tend to use AI while simultaneously being more productive by default?

Validity

This particular dimension of rigor can be framed as **internal validity**.

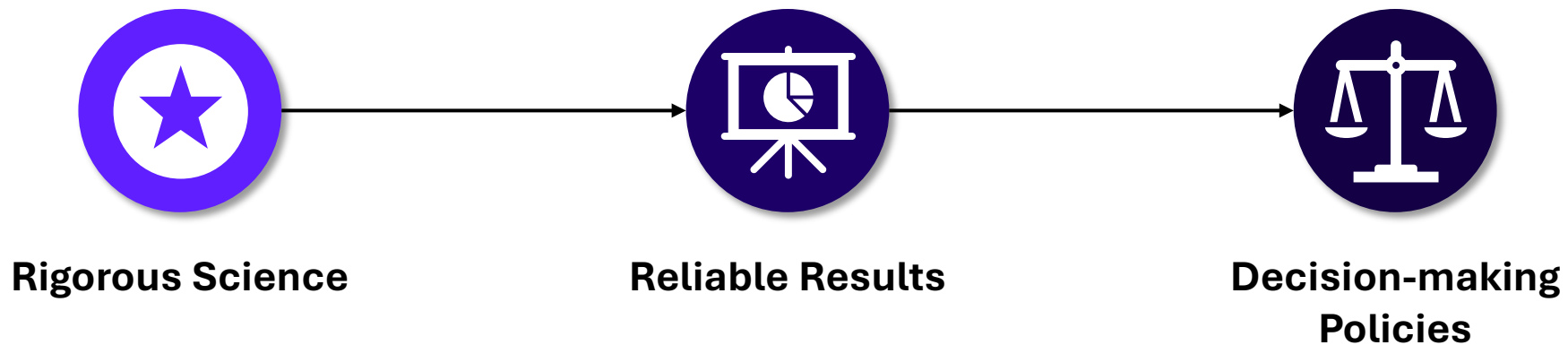


Wohlin, C., Runeson, P., Höst, M., Ohlsson, M. C., Regnell, B., & Wesslén, A. (2012). Experimentation in software engineering (Vol. 236, p. P3). Berlin: Springer.

Internal validity is the degree to which an observed effect can be attributed to the independent variable of interest.

The Scientific Process

Distinguishing **causal relationships** from **spurious relationships** is why our research needs to be *rigorous*. Otherwise, our results are not reliable and do not serve decision-making policies.



We need tools to frame and identify this distinction.



Causal Modelling

A Syntax to Formalize Causal Assumptions

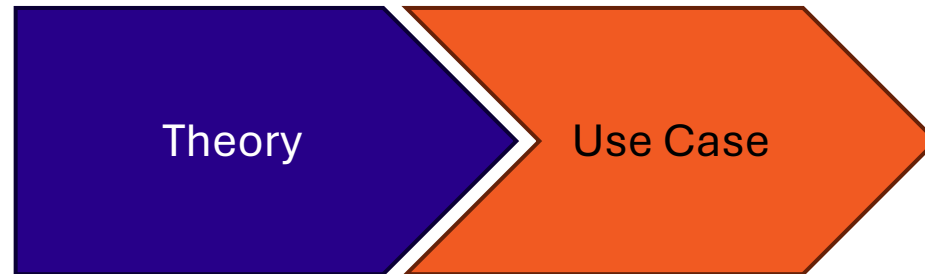
Causal Modelling

A (structural) causal model is a **conceptual model** that *represents the causal mechanisms of a system* [1]. They serve two main purposes [2]:

1. **Visualizing causal assumptions** and determining testable implications.
2. **Proving the possibility** of computing causal effects from observational data.

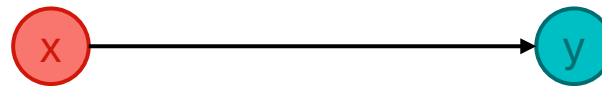
Structure of the Slides

Every lesson consists of its theory and one or more use cases.



Directed Acyclic Graphs

Causal models can be realized as directed, acyclic graphs (DAGs).



exposure



response



(assumed) causal
relationship



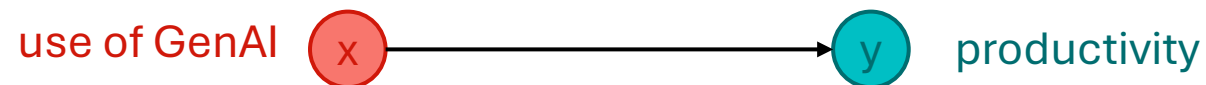
phenomenon

Formulating Casual Hypotheses

DAGs can visualize a causal hypothesis graphically

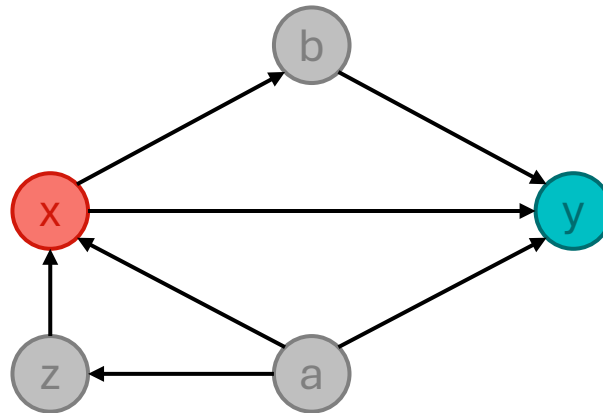
”

RQ: Does the use of **generative AI (GenAI)** have an **effect** on the **productivity** of developers?



Directed Acyclic Graphs

Causal models can be realized as directed, acyclic graphs (DAGs).



exposure



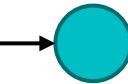
response



other factors



(assumed) causal
relationship



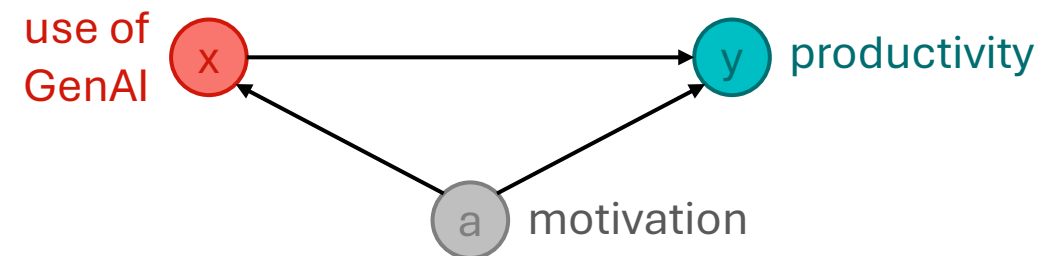
phenomenon

Formulating Complex Hypotheses

DAGs convey complex relationships between multiple variables in a simple and straightforward manner.

”

RQ: Does the use of **generative AI (GenAI)** have **an effect** on the **productivity** of developers, assuming that **motivation** affects both usage and productivity.



Formulating Complex Hypotheses

DAGs convey complex relationships between multiple variables in a simple and straightforward manner.

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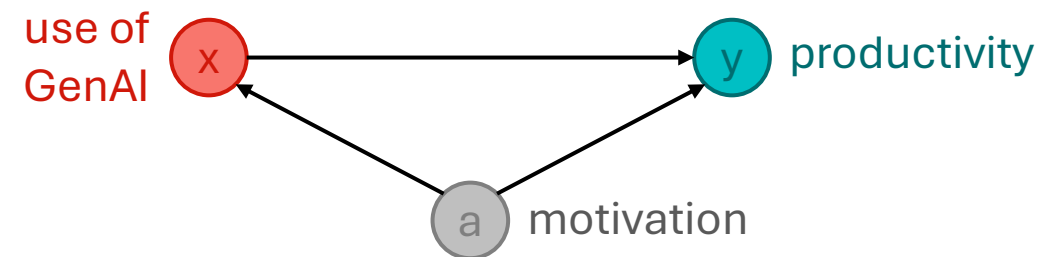
RQ: Does the use of **generative AI (GenAI)** have **an effect** on the **productivity** of developers, assuming that motivation affects both usage (mediated by GenAI experience) and productivity, and accounting for an indirect effect of the IDE's user experience?

Formulating Complex Hypotheses

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RQ: Does the use of **generative AI (GenAI)** have **an effect** on the **productivity** of developers, assuming that **motivation** affects both usage (mediated by GenAI experience) and productivity, and accounting for an indirect effect of the IDE's user experience?

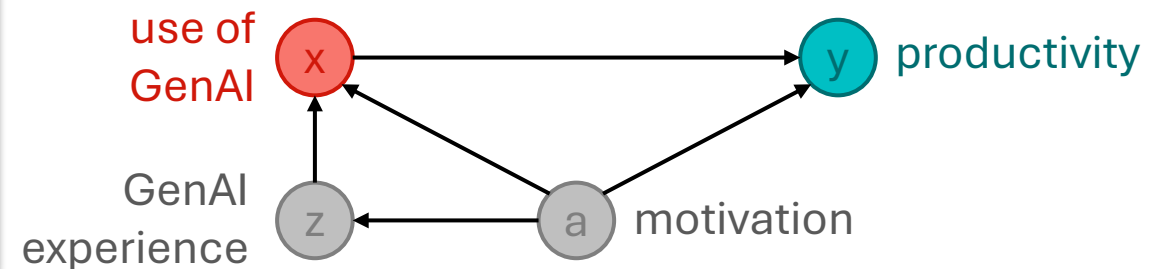


Formulating Complex Hypotheses

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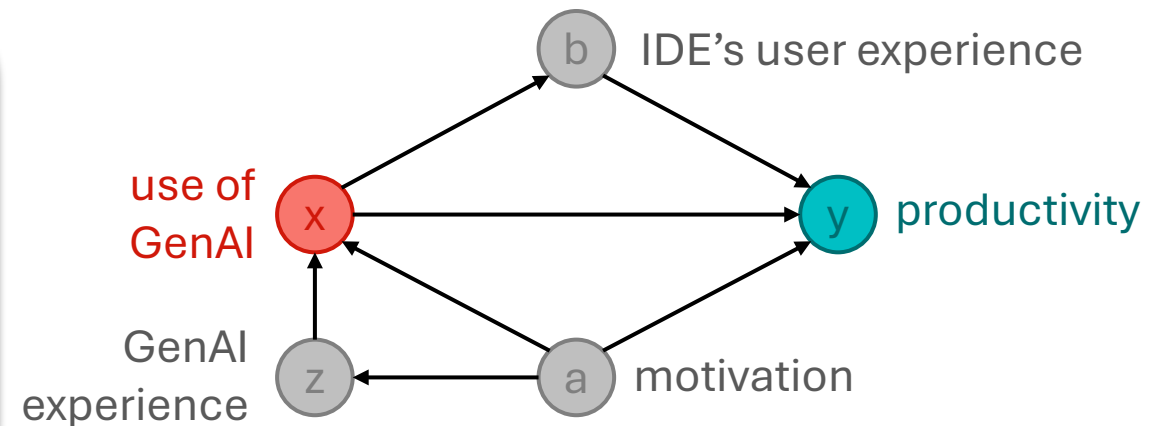


Formulating Complex Hypotheses

DAGs convey complex relationships between multiple variables in a simple and straightforward manner.

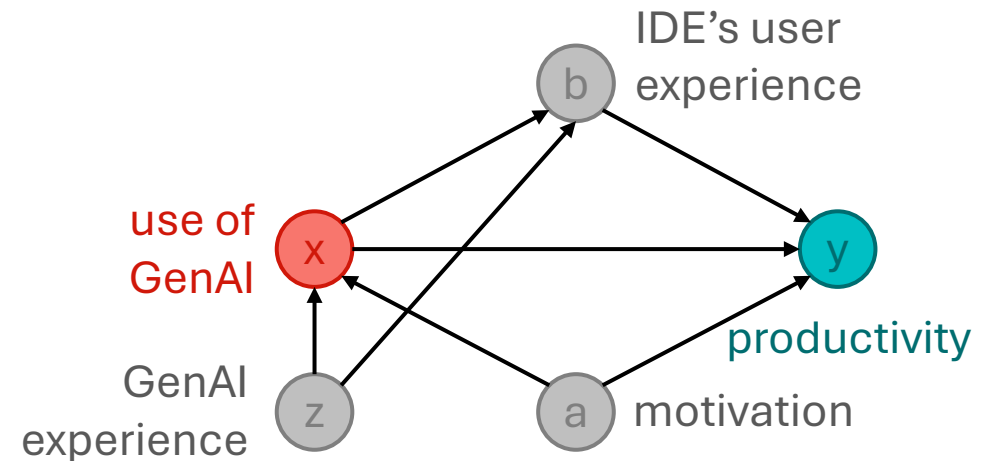
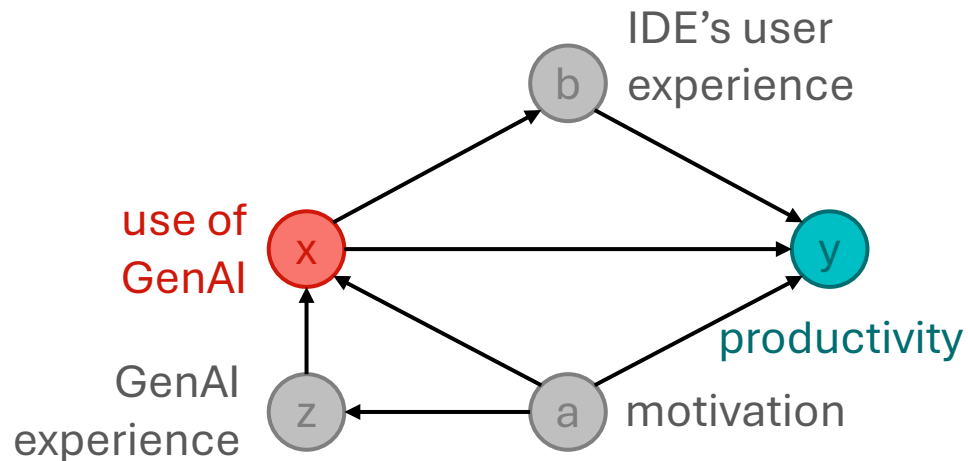
”

RQ: Does the use of **generative AI (GenAI)** have **an effect** on the **productivity** of developers, assuming that **motivation** affects both usage (mediated by **GenAI experience**) and productivity, and accounting for an indirect effect of the **IDE's user experience**?



Disputing Causal Hypotheses

The explicit formulation of causal hypotheses in DAGs makes it easy to compare and contrast assumptions.

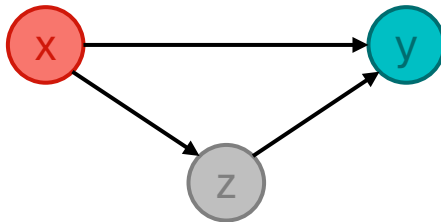


These comparisons can inform the design of an empirical study to verify, which of the two hypotheses reflects reality better.

Types of Relationships

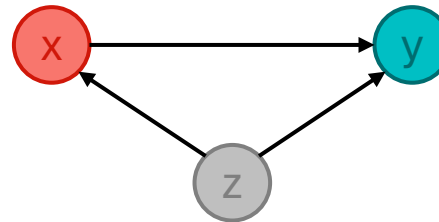
When a DAG involves more than two variables, certain relationships can emerge:

Mediation



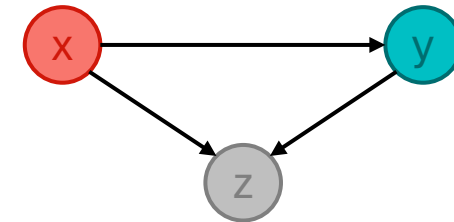
z *mediates* the effect of x on y . There is a direct ($x \rightarrow y$) and an indirect ($x \rightarrow z \rightarrow y$) effect.

Confounding



z *confounds* the effect of x on y . An observed effect ($x \rightarrow y$) might be caused by the confounder ($x \leftarrow z \rightarrow y$).

Collider

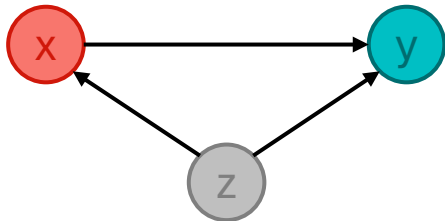


z is a common effect of x and y .

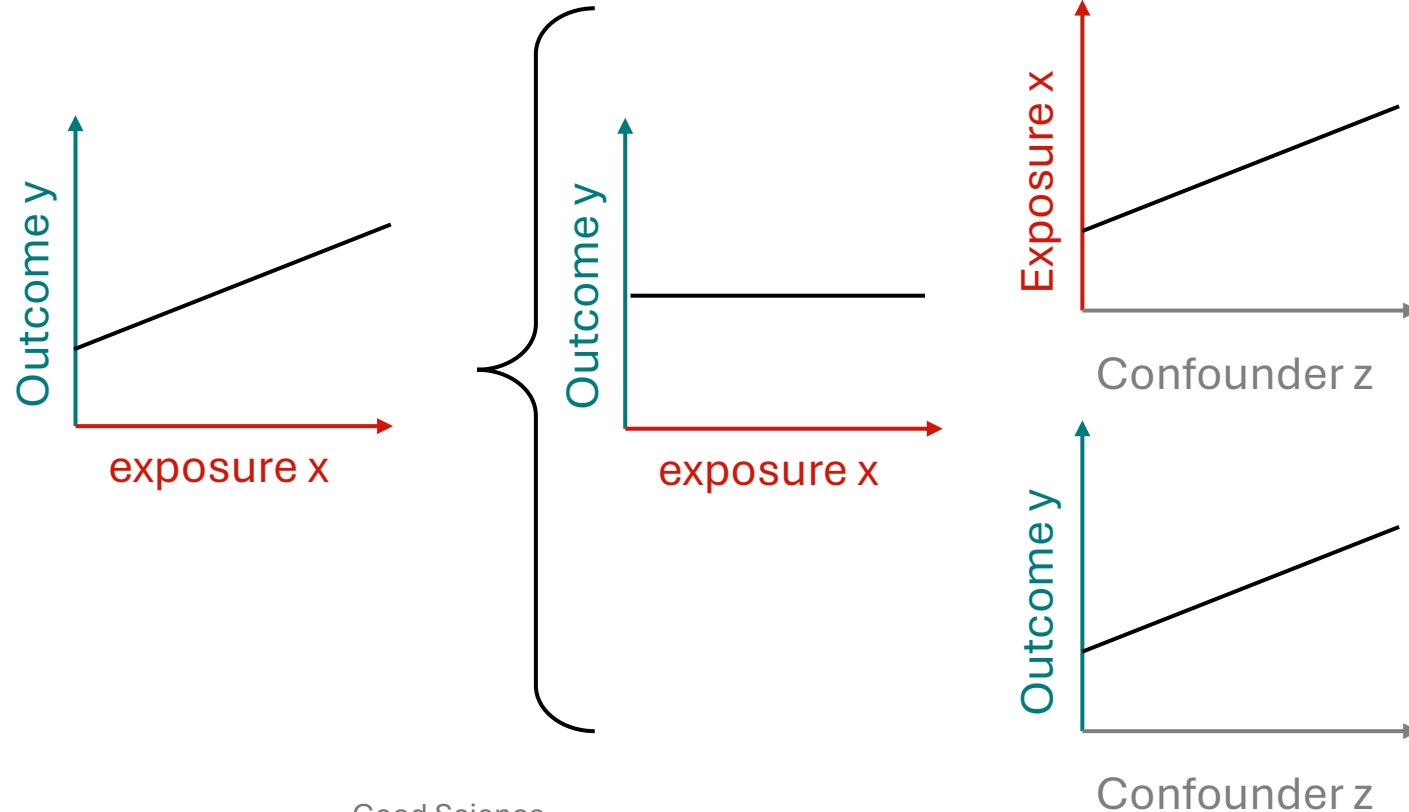
The Effect of The Third Variable

How does a third variable affect the relationship of interest?

Confounding



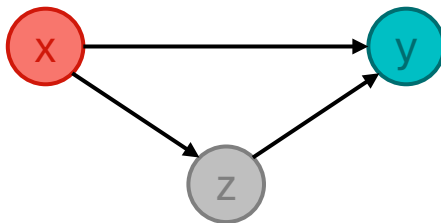
z *confounds* the effect of x on y . An observed effect ($x \rightarrow y$) might be caused by the confounder ($x \leftarrow z \rightarrow y$).



Dealing with Variables

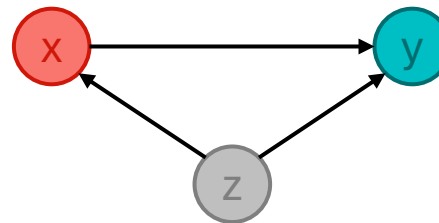
Based on the type of relationship between variables, these additional factors need to be treated differently.

Mediation



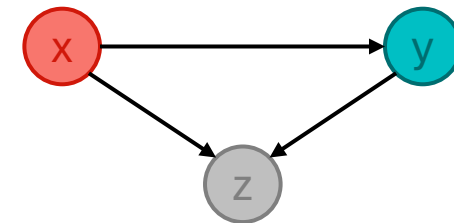
You **can** control z if you want to distinguish the direct from the indirect effect

Confounding



You **must** control z to avoid confounding

Collider



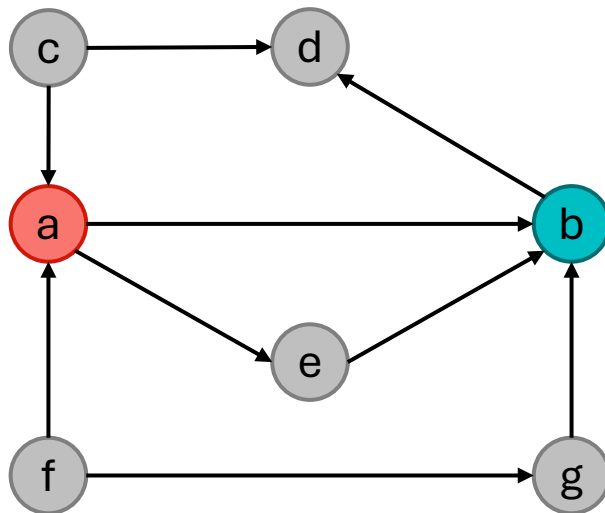
You **must not** control z to avoid confounding

Control can either be by design (fixing variables) or by analysis (stratifying the data).

Identifying Paths

Understanding the influence of additional variables on a relationship of interest (**exposure** → **outcome**) allows us to **de-confound this relationship** and *draw causal conclusions*.

For this, we need to understand **paths**. A path is any non-cyclic, consecutive sequence of relationships in a DAG.



closed path: path that contains a collider (d)

(open) backdoor path: path that starts with an arrow entering the **exposure**

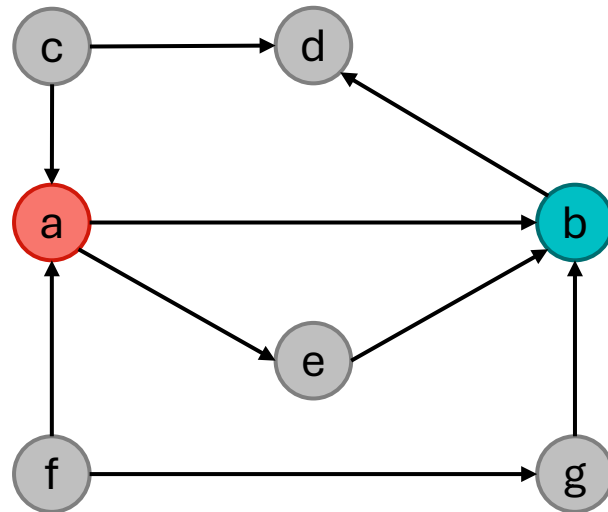
d-Separation

A general framework to derive statistical models from causal models uses **d-separation**. It produces an **adjustment set**, i.e., the set of variables to control statistically in order to de-confound the causal effect between the exposure and the outcome.

<https://bayes.cs.ucla.edu/BOOK-2K/d-sep.html>

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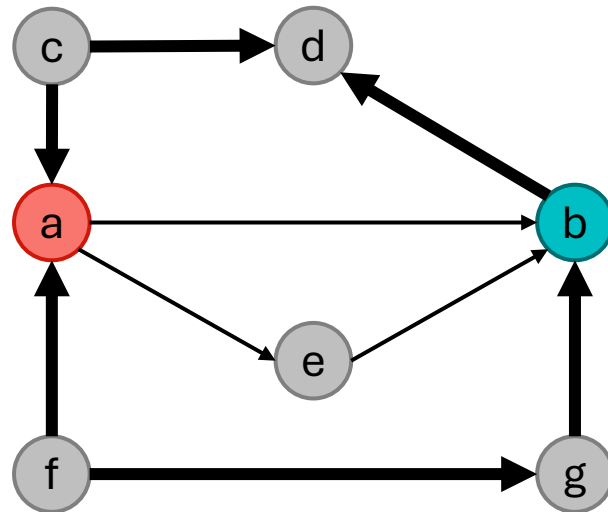
Model Derivation

Adjustment set = {a}

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Model Derivation

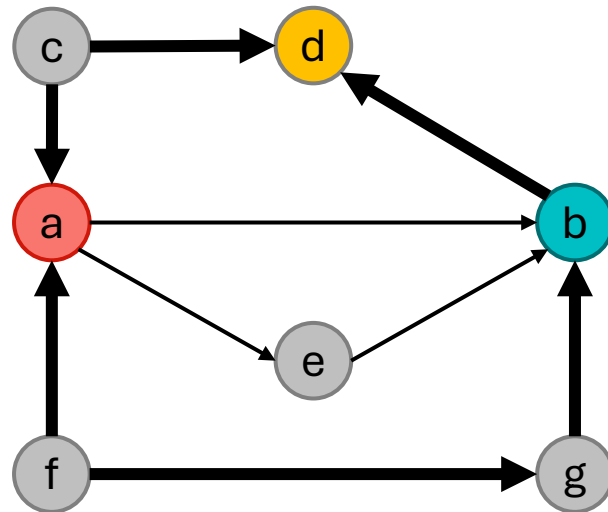
Adjustment set = {**a**}

1. Identify every **backdoor path**.

<https://bayes.cs.ucla.edu/BOOK-2K/d-sep.html>

d-Separation

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Model Derivation

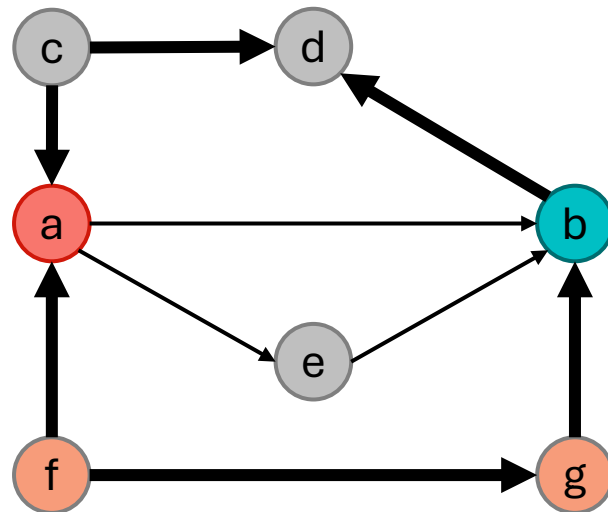
Adjustment set = {**a**}

1. Identify every backdoor path.
2. Check if there is a **collider** between the cause and effect of interest.
 1. If yes, do not condition on any of the variables on this path.

<https://bayes.cs.ucla.edu/BOOK-2K/d-sep.html>

d-Separation

A general framework to derive statistical models from causal models uses **d-separation**. It produces an **adjustment set**, i.e., the set of variables to control statistically in order to de-confound the causal effect between the exposure and the outcome.



Model Derivation

Adjustment set = {a, f}

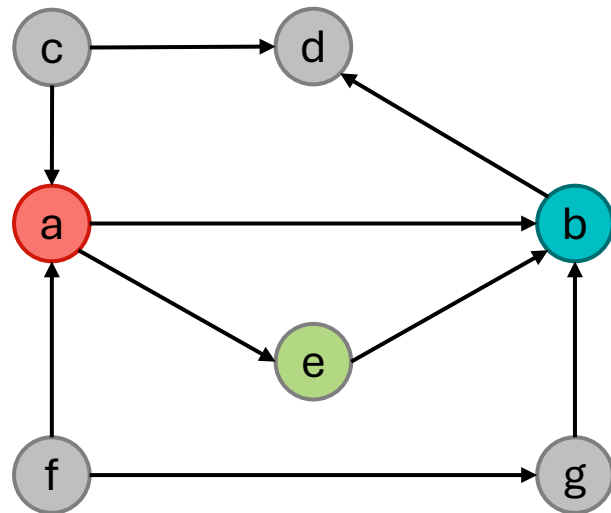
Adjustment set = {a, g}

1. Identify every backdoor path.
2. Check if there is a collider between the cause and effect of interest.
 1. If yes, do not condition on any of the variables on this path.
 2. If not, condition on **any of the variables** on this path to “close” the backdoor path.

<https://bayes.cs.ucla.edu/BOOK-2K/d-sep.html>

d-Separation

A general framework to derive statistical models from causal models uses **d-separation**. It produces an **adjustment set**, i.e., the set of variables to control statistically in order to de-confound the causal effect between the exposure and the outcome.



Model Derivation

Adjustment set = {a, f, [e]}

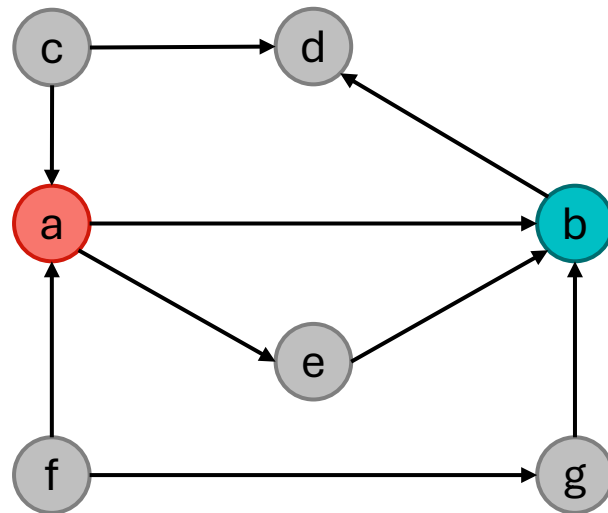
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3. Optionally, condition on **mediators** to discern the direct effect.

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d-Separation

A general framework to derive statistical models from causal models uses **d-separation**. It produces an **adjustment set**, i.e., the set of variables to control statistically in order to de-confound the causal effect between the exposure and the outcome.



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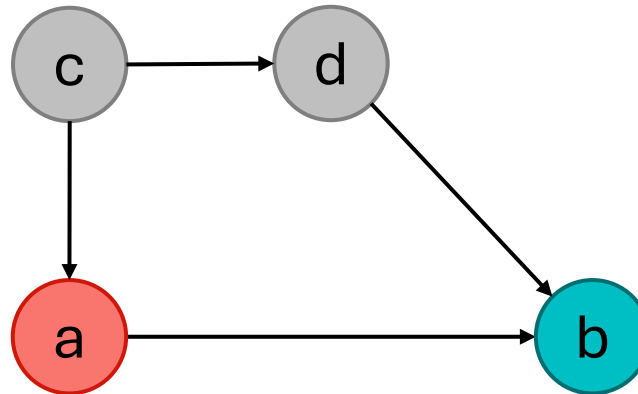
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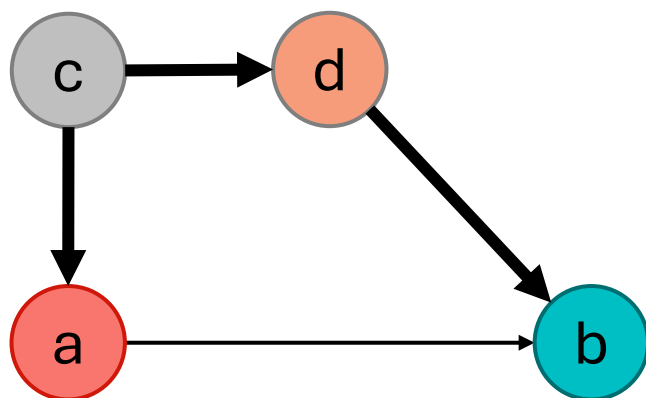
Exercises

Determine the **adjustment set** of variables to de-confound the estimation of the effect of the phenomenon $a \rightarrow b$ via d-separation.

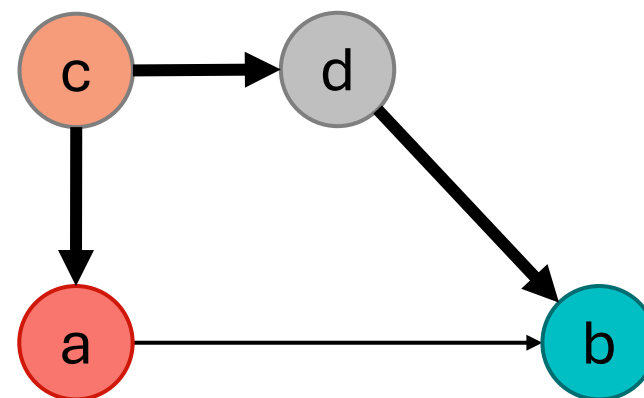


Exercises

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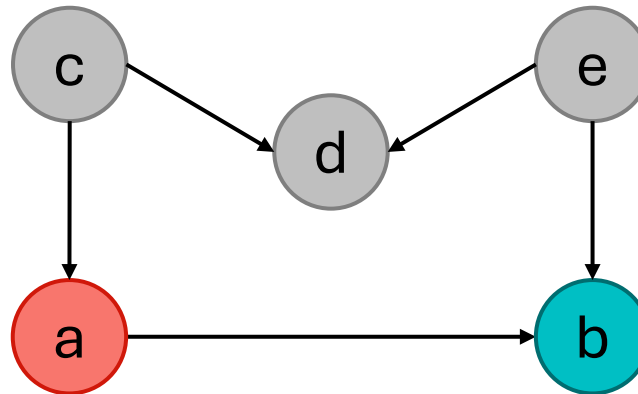
Adjustment set = {d}



Adjustment set = {c}

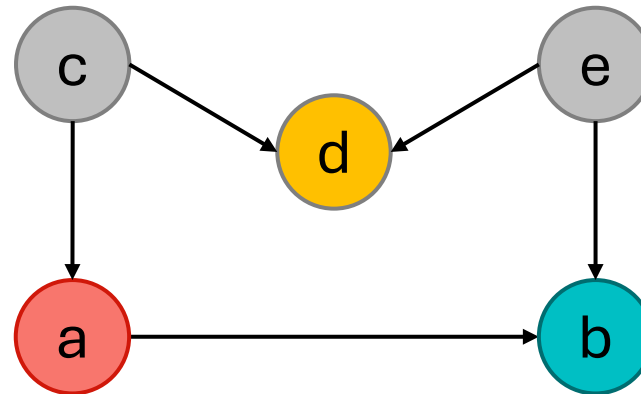
Exercises

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Exercises

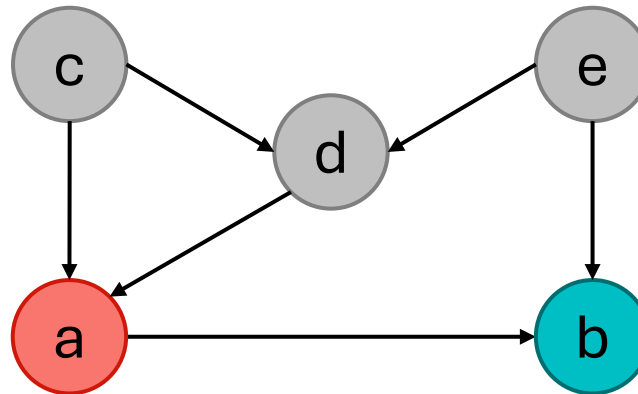
Determine the **adjustment set** of variables to de-confound the estimation of the effect of the phenomenon $a \rightarrow b$ via d-separation.



Adjustment set = $\{\}$

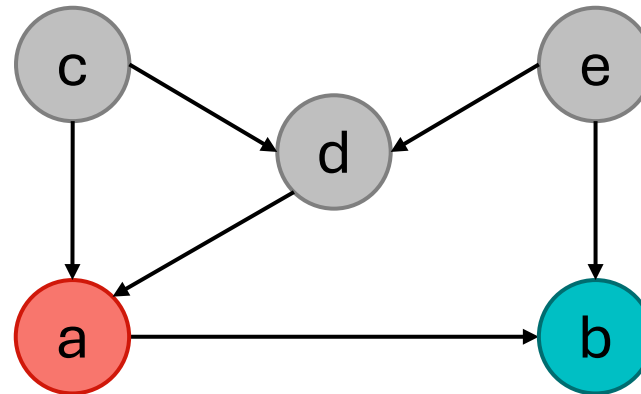
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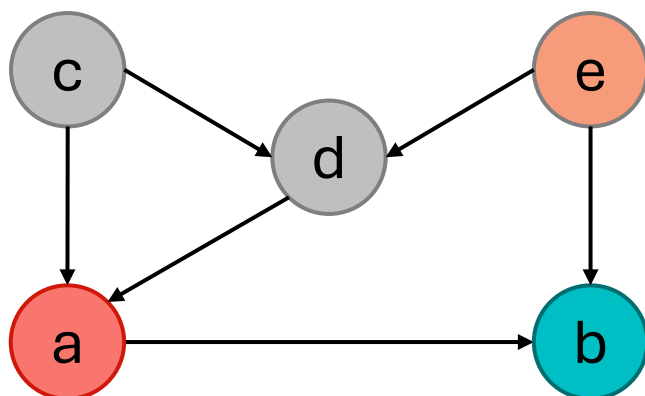


The DAG contains two backdoor paths:

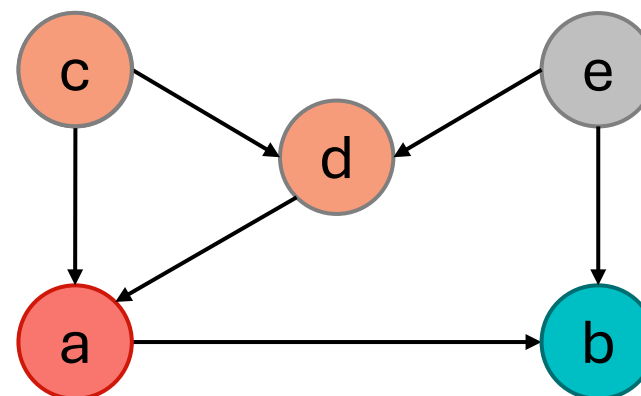
1. $a \leftarrow c \rightarrow d \leftarrow e \rightarrow b$
2. $a \leftarrow d \leftarrow e \rightarrow b$

Exercises

Determine the **adjustment set** of variables to de-confound the estimation of the effect of the phenomenon $a \rightarrow b$ via d-separation.



Adjustment set = {e}



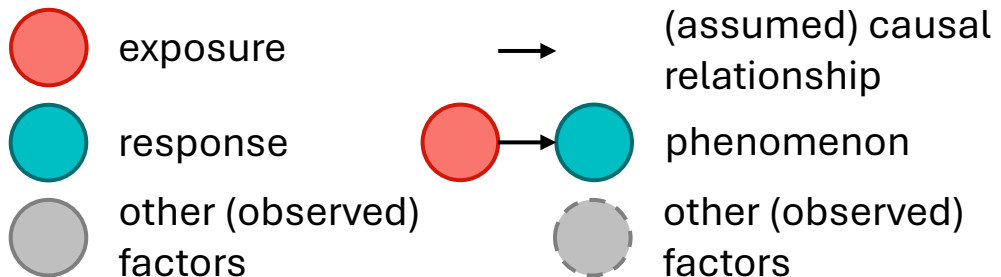
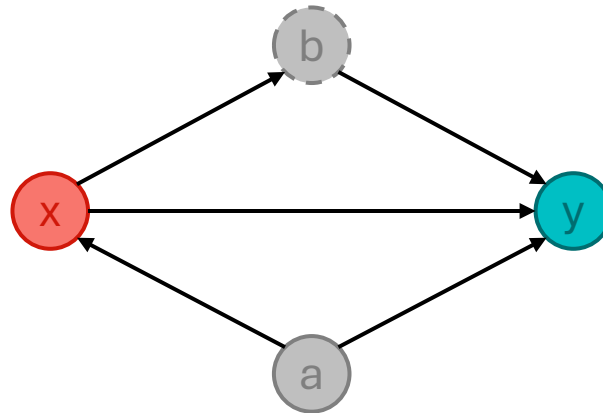
Adjustment set = {c, d}

”

Conditioning on d opens the backdoor-path $a \leftarrow c \rightarrow d \leftarrow e \rightarrow b$ which then has to be closed by additionally conditioning on c .

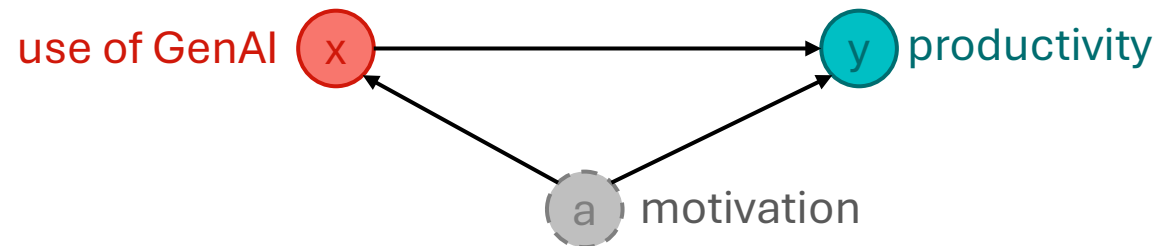
Encoding Observability

Nodes in DAGs can indicate whether a variable is observable or not.



Determining Study-ability

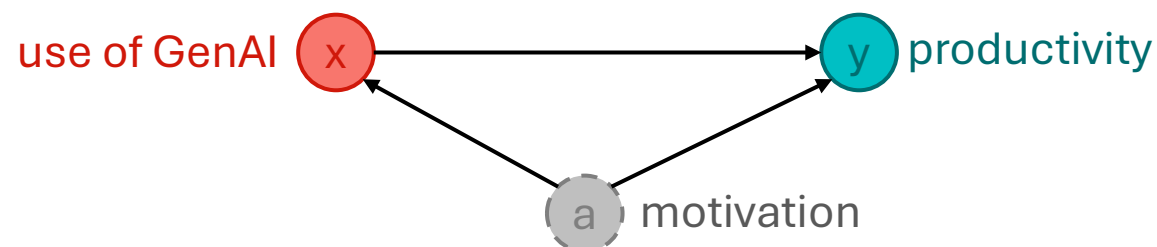
Causal DAGs allow deciding whether it is even possible to determine the (unbiased) causal effect of a phenomenon.



If a phenomenon has an unobserved confounder, then the causal effect of the exposure on the outcome cannot be estimated from observational data.

Limitations of Observational Studies

Unobservable/unmeasurable or even unknown confounders limit the potential of observational studies.



In such a case, we would only be able to detect a **spurious association** between the use of GenAI and productivity, but never the **true causal** effect.



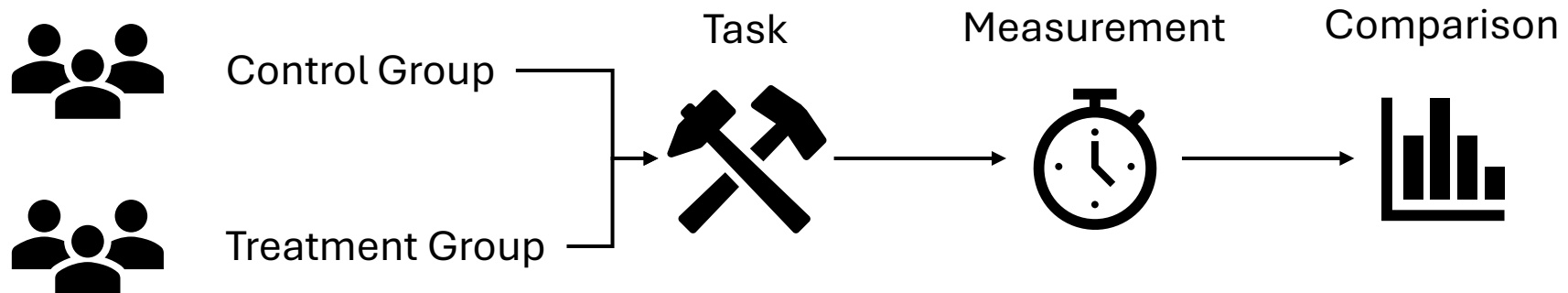
Experimentation

Controlling Away Confounding

Controlled Experiments

” A controlled experiment is a research method defined by its **random treatment** (=exposure) **assignment strategy**.

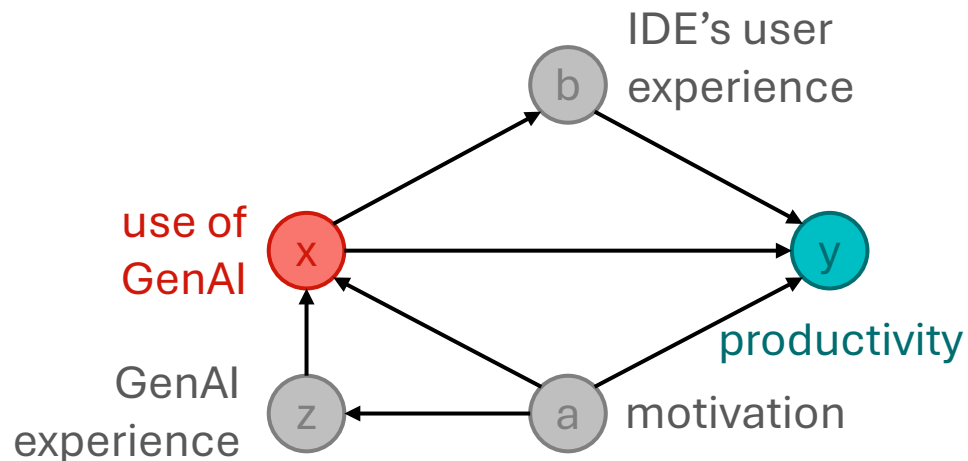
You as the experimenter exert **control** by assigning the (levels of the) exposure to the subjects of the study yourself.



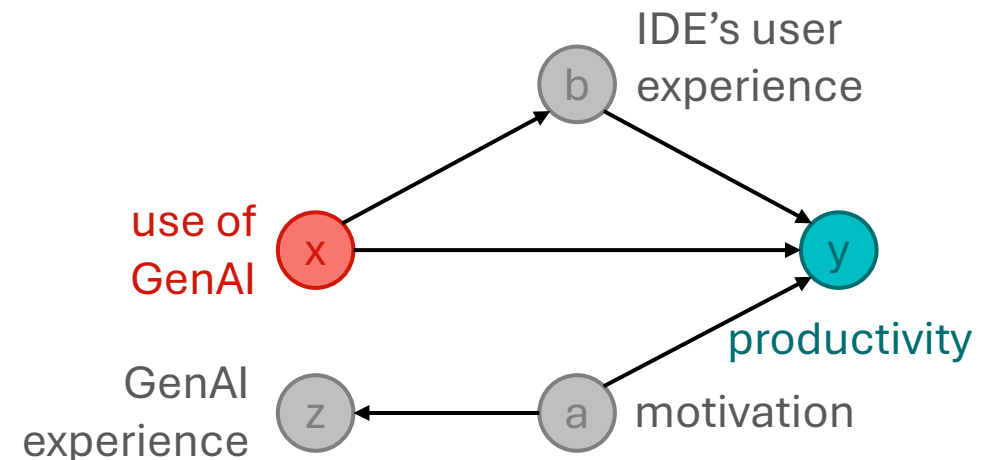
Distinguishing Study Types

The causal DAG also reveals controlled vs. observational studies: a controlled study is indicated by a DAG where no variable affects the exposure.

Observational study



Controlled Experiment



Controlled Experiments in Software Engineering

Software engineers use controlled experiments at several occasions, for example:

- A/B testing versions of a user interface
- Comparing configurations of ML models
- ...

The internal validity of controlled experiments makes them very **reliable sources of causal insights**. So why are we not using experiments all of the time?

Limitations

There are several reasons why we cannot investigate every phenomenon with a controlled experiment, as they are



Expensive

Recruiting eligible participants is difficult



Unrealistic

Experiments perturb the natural context

Threats to External Validity



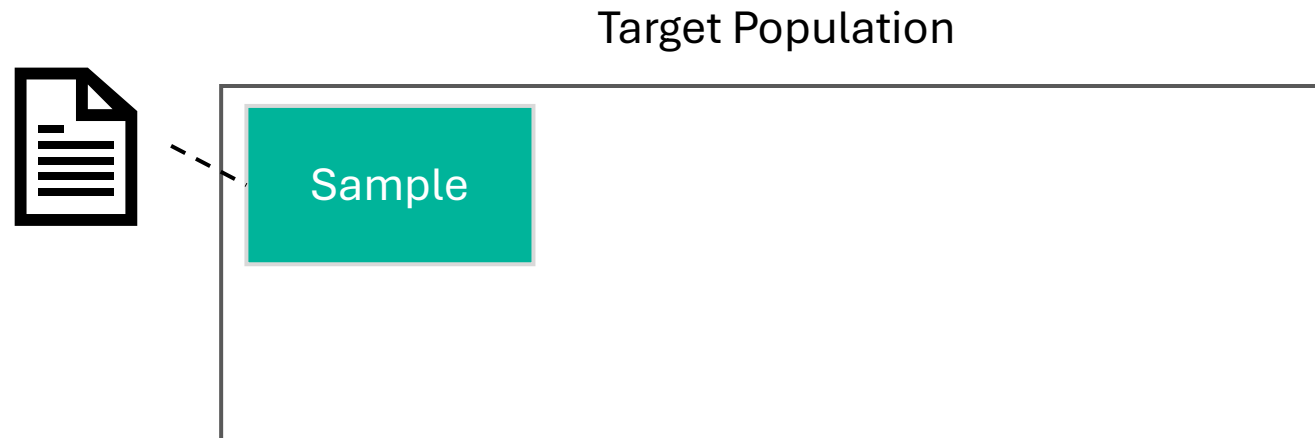
Wohlin, C., Runeson, P., Höst, M., Ohlsson, M. C., Regnell, B., & Wesslén, A. (2012). Experimentation in software engineering (Vol. 236, p. P3). Berlin: Springer.

External validity is “our ability to generalize the results of our experiment to industrial practice”

External validity determines how useful results from a study are outside of its scope.

Target Population and Representative Samples

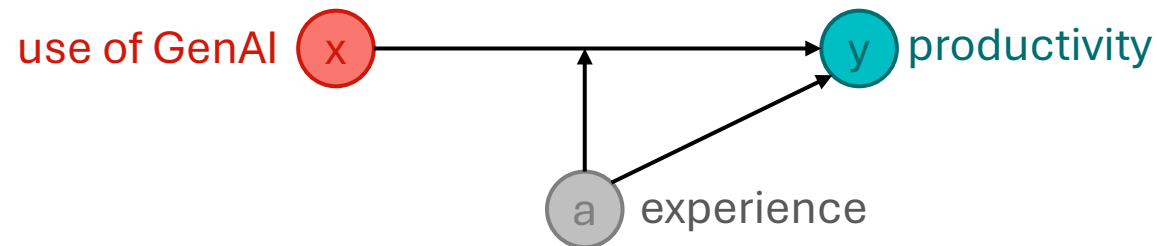
It is impossible to study *all* subjects and objects of interest (i.e., the target population), so we need to **draw a (representative) sample** from it.



If the sample is *representative*, we can still infer about the target population, which is our goal.

Moderation

Factors can not only affect other factors, but also other relationships in a process called moderation.

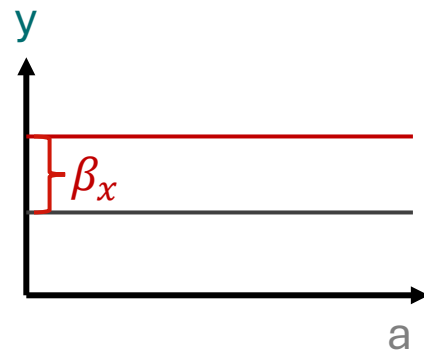
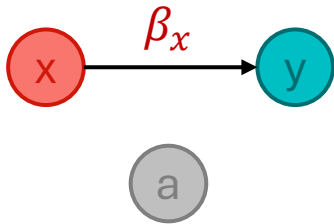


A moderator changes the effect that one variable has on another.

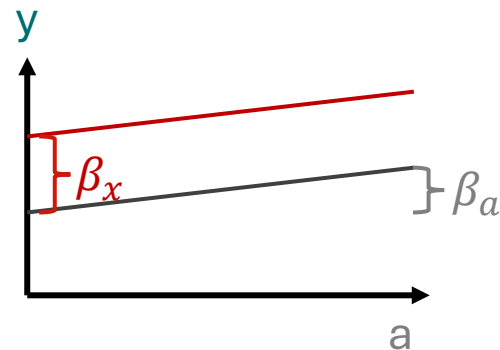
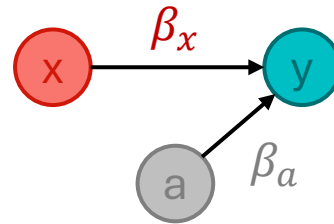
Determining Moderation

The existence of a moderator evokes a distinct effect pattern.

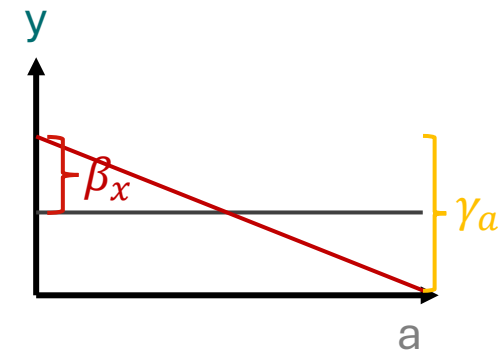
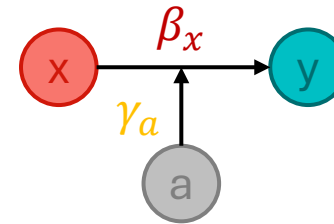
no covariation,
no moderation



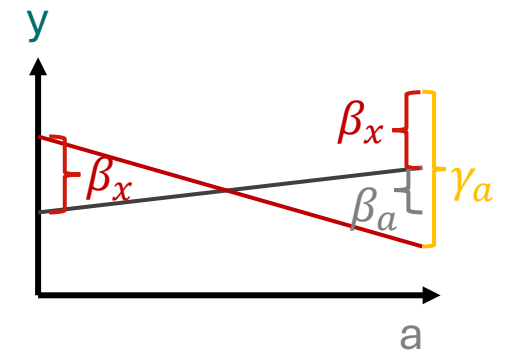
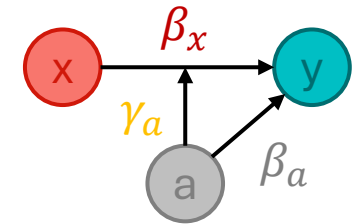
covariation, but
no moderation



no covariation,
but moderation



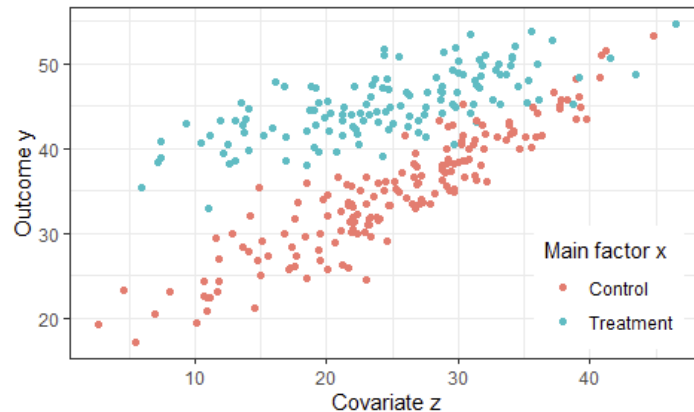
covariation and
moderation



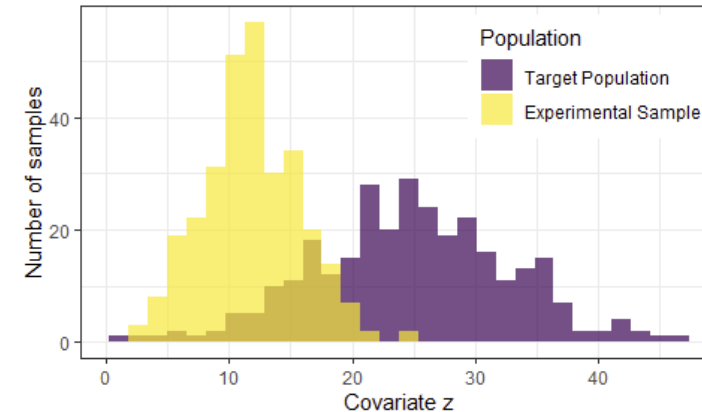
Threat to External Validity

When studying a phenomenon via a controlled experiment and...

(1) the average treatment effect is **moderated** by a covariate,



(2) This covariate's **distribution differs** in the experimental sample from the target population



... then the external validity of the result is threatened.

Limitations

There are several (more) reasons why we cannot investigate every phenomenon with a controlled experiment, as they are



Expensive

Recruiting eligible participants is difficult



Unrealistic

Experiments perturb the natural context



Impossible

Some factors are impossible to control



Assumed

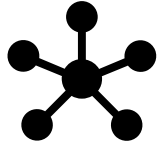
The validity of experimental results depends on several conditions



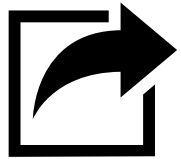
Conclusion

What Does This Mean for (Empirical) Software Engineering

How to Good Science



Internal validity requires visualizing and communicating your causal assumptions, and closing any backdoor paths by controlling confounders



External validity requires awareness of potential moderators that also affect trial eligibility